# A Synopsis
## On
# CNN VIT BILSTM CROSS-ATTENTION BASED TRAFFIC MANAGEMENT SYSTEM

Submitted in partial fulfilment of the requirement for the award of the degree of

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**IN**
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# CERTIFICATE

This is to certify that the project synopsis entitled "CNN VIT BILSTM CROSS-ATTENTION BASED TRAFFIC MANAGEMENT SYSTEM" is being submitted by Divyansh Uniyal (2318766), Bhavya Gogia (2318662), Km Shivani (2318999), Abhishek Negi (2318178) for the partial fulfillment of the requirement for the award of the degree of B.Tech C.S.E. (Machine Learning & Artificial Intelligence) in Department of Computer Science and Engineering, Graphic Era Hill University, Dehradun, and is a record of their own work carried out under my supervision. The matter embodied in the project synopsis has not been submitted in any other university or institution for the award of any degree.

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# CANDIDATE'S DECLARATION

We hereby certify that the work which is being presented in the project synopsis entitled "CNN VIT BILSTM CROSS-ATTENTION BASED TRAFFIC MANAGEMENT SYSTEM" is being submitted for the partial fulfillment of the requirement for the award of the degree of B.Tech C.S.E. (Machine Learning & Artificial Intelligence) in Department of Computer Science and Engineering, Graphic Era Hill University, Dehradun, is our original work and has not been submitted to any other university or institution for the award of any degree or diploma.

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# ABSTRACT

Urban traffic in India is heterogeneous and only weakly lane disciplined: two-wheelers filter between larger vehicles, auto-rickshaws and e-rickshaws occupy a lane share that no vehicle-count threshold captures, and fixed-time signal plans commissioned from a historical survey continue to run long after the demand that justified them has changed. Loop detectors and radar are the conventional remedy, and their per-intersection cost keeps them out of reach for most municipal deployments. Cameras, by contrast, are already installed.

This project proposes MFSTNet, a camera-only system that forecasts per-lane congestion 60 seconds ahead and hands that forecast to a reinforcement-learning signal controller. A fine-tuned YOLOv8s detector, trained over eight vehicle classes that include the auto-rickshaw, e-rickshaw and cattle absent from general-purpose datasets, produces per-lane counts. Frozen ResNet-50 and DINOv2 ViT-S/14 backbones supply complementary appearance features, which a gated bidirectional cross-attention module fuses into a single representation; the gate value is retained and reported as a research artefact rather than discarded as an internal detail. A BiLSTM with temporal self-attention models the 60-step observation window, and a per-lane head classifies each approach as LOW, MEDIUM or HIGH. A Proximal Policy Optimization agent then consumes these predictions inside a 16-dimensional observation vector and selects signal phases in a SUMO microsimulation of a four-way intersection.

The methodology is deliberately falsifiable. Every architectural component is disableable by configuration, giving an eight-configuration ablation that includes a linear probe as its floor; evaluation is leave-one-camera-out, so no reported score can be a memorised camera; and all results are reported as means with bootstrap 95% confidence intervals across folds and seeds. On the 4-fold, 5-seed pilot corpus the strongest vision configuration reaches a macro-F1 of 0.4405, while a count-sequence baseline (gru counts) reaches 0.5663; the vision claim is therefore reported as open rather than established. On the control side, the trained policy reduces mean vehicle waiting time by 19.35% against a swept, best-configuration Webster baseline over 30 paired seeds ($p < 0.0001$, Cohen's $d = -1.34$).

The expected outcome is a reproducible, licence-audited dataset, a trained and exported model, a complete ablation and control benchmark with raw per-run results committed alongside the code, and an edge-to-server prototype in which no raw video frame ever leaves the capture device.

**Keywords:** traffic congestion forecasting, multimodal fusion, vision transformer, cross-attention, reinforcement learning, adaptive signal control, SUMO, edge inference.

# CHAPTER 1

# INTRODUCTION

## 1.1 Background

A signalised intersection allocates a scarce resource — green time — among competing approaches. The allocation rule in near-universal use in Indian cities is fixed-time control, in which cycle length and phase splits are computed once from a manual survey and then held constant. Webster's 1958 formulation remains the reference method for that computation and is still the honest baseline any adaptive scheme must beat, because it is what is actually deployed.

Adaptive control instead measures demand and reallocates green time in response to it. The measurement is the difficulty. Inductive loops must be cut into the carriageway and fail with the road surface; radar and lidar units are priced per approach. Both assume that vehicles arrive in lanes, which in unstructured traffic they do not: a stream of filtering two-wheelers registers as a small count while occupying the discharge capacity of a car.

Cameras avoid both problems. They are already mounted at a large fraction of Indian urban intersections for enforcement and surveillance, they observe the whole approach rather than a point, and modern detectors can be run on commodity hardware near the camera. What a camera does not provide directly is the quantity a controller needs, which is not the present state of the intersection but its state a minute from now — long enough for a phase decision made now to still be the right one when it takes effect.

## 1.2 Motivation

Three observations motivate this work. First, a controller that reacts to present queues is always one decision behind the traffic; a controller supplied with a short-horizon forecast can act before a queue forms. Second, general-purpose detectors and datasets do not represent Indian road users — COCO contains no auto-rickshaw and no e-rickshaw class, and a detector that cannot name a vehicle cannot count it. Third, the published literature on multimodal fusion for traffic prediction reports architecture comparisons without the floor that a frozen-backbone study owes its reader: a linear probe on the same features. Where that floor is missing, an architectural gain and a feature-quality gain are indistinguishable.

This project therefore treats the ablation, not the architecture, as the deliverable of scientific interest. Every module is disableable by a configuration flag, the linear probe is a first-class

configuration rather than an afterthought, and a negative or inconclusive result is reported and analysed rather than dropped.

## 1.3 Significance and Need

The direct beneficiary is the municipal traffic authority that already owns camera infrastructure and cannot fund a sensor retrofit. The system requires no new roadside hardware, produces a control decision in under 50 ms, and degrades safely: if the forecast is unavailable the controller runs on raw counts, and if the network is lost for more than 10 seconds the edge device falls back to a locally embedded Webster controller. Neither fallback is an optional extra; both are specified behaviour and both are tested.

The design also takes privacy seriously as a constraint rather than a disclaimer. Frames are processed where they are captured and are never transmitted over the network or written to disk; only derived counts and predictions leave the edge device. A deployment therefore cannot become a video archive by accident.

## 1.4 Organisation of the Synopsis

Chapter 2 reviews vehicle detection in unstructured traffic, spatio-temporal congestion prediction, multimodal fusion and reinforcement-learning signal control, and identifies the gap this work addresses. Chapter 3 states the problem precisely and lists the objectives. Chapter 4 describes the proposed methodology, including the system architecture, the corpus construction pipeline, the model, the controller, the evaluation protocol and the preliminary results obtained so far. Chapter 5 lists the hardware and software requirements, Chapter 6 the expected outcomes, and Chapter 7 concludes.

# CHAPTER 2

# LITERATURE REVIEW

## 2.1 Vehicle Detection in Unstructured Traffic

Single-stage detectors dominate deployed traffic sensing because they meet real-time budgets on modest hardware. The YOLO family [5] is the usual choice, and its accuracy on Western traffic imagery is well established. The limitation is categorical rather than architectural: models pre-trained on COCO recognise car, motorcycle, bus and truck, and have no label for the auto-rickshaw, e-rickshaw, or cattle that occupy Indian carriageways. The India Driving Dataset [6] was assembled specifically to expose this, and demonstrates that models trained on structured-traffic imagery degrade sharply on unconstrained Indian roads. Most such datasets are nonetheless captured from a dashcam viewpoint, whereas an intersection controller sees an elevated, fixed CCTV view — a domain shift that the literature reports but rarely quantifies.

## 2.2 Spatio-Temporal Congestion Prediction

Congestion forecasting is conventionally posed over sensor time series. Recurrent models [14] capture the temporal dependency, and gradient-boosted trees [15] remain a strong tabular baseline. Both consume counts or occupancies, which presupposes an instrumented network. Vision-based forecasting instead reads congestion from the image, which removes that presupposition but introduces another: appearance features must carry information about the *future* state, and whether they do is an empirical question that much of the literature assumes rather than tests. A count-only sequence model on the same footage is the control condition that settles it, and it is reported far less often than it should be.

## 2.3 Multimodal Fusion of Convolutional and Transformer Features

Convolutional backbones [4] encode local texture with strong inductive bias; vision transformers [2] encode long-range relationships through self-attention [1]. Because the two are complementary, fusing them is a natural design, and the mechanisms for doing so — concatenation, one-directional cross-attention, bidirectional cross-attention, and gated combination — are all individually published. Self-supervised pre-training [3] has since changed the calculus: DINOv2 features transfer well enough that freezing both backbones and training only the fusion layers is a defensible design, which in turn makes the linear

probe the mandatory floor. If a linear classifier on the same frozen features matches the fusion network, the fusion network is unjustified, and that is a result rather than a failure.

## 2.4 Reinforcement Learning for Signal Control

Reinforcement learning for traffic signal control is a mature subfield; the survey of Wei et al. [13] catalogues both its progress and its recurring methodological weaknesses, chief among them baselines that are weaker than what is deployed and evaluations reported without variance. Proximal Policy Optimization [7] is the standard on-policy algorithm and is available in a maintained reference implementation [8]. SUMO [9] is the usual simulator; modelling non-lane-disciplined traffic requires its sublane extension, since the default model forbids the lateral filtering that defines the regime. On the baseline side, Webster's method [10] is sensitive to an assumed saturation flow whose published values for heterogeneous traffic span more than a factor of two [11], [12] — so quoting a single value silently chooses how strong the baseline will be.

## 2.5 Comparative Summary

Table 2.1 places the four strands beside one another. The pattern is that each strand is individually well developed and the combination is not: detection work rarely carries through to control, forecasting work rarely reports the count-only control condition, fusion work rarely reports the linear-probe floor, and control work rarely sweeps its baseline.

**Table 2.1 Comparison of the four relevant research strands**

| Strand | Representative work | Strength | Limitation for this problem |
|---|---|---|---|
| Detection | YOLO family [5]; IDD [6] | Real-time on modest hardware; India-specific classes exist | Dashcam viewpoint dominates; CCTV-view performance seldom reported |
| Forecasting | LSTM [14]; XGBoost [15] | Strong on sensor time series | Assumes instrumented network; count-only control condition often omitted |
| Fusion | ViT [2]; DINOv2 [3]; cross-attention [1] | Complementary local and global features; frozen backbones viable | Linear-probe floor rarely reported, so gains are unattributable |
| RL control | PPO [7]; SB3 [8]; SUMO [9] | Mature algorithms and simulators | Weak or unswept baselines [10]- |

| Strand | Representative work | Strength | Limitation for this problem |
|---|---|---|---|
| | | | [12]; variance often unreported |

## 2.6 Research Gap

Three gaps follow, and the project is scoped to them rather than to a claim of architectural novelty, which the fusion literature would not support.

**G1.** Camera-only, forecast-driven control. Detection, forecasting and control are studied separately; an end-to-end path from a single existing camera to a signal decision, with no additional roadside sensing, is not established for unstructured traffic.

**G2.** The gate as a reported artefact. Gated fusion is published as a mechanism, but the gate value is treated as an internal activation. Exposing it — logging it per run, tracking it live, and analysing when it saturates — turns it into evidence about which modality the model is actually using.

**G3.** Density-stratified, leave-one-camera-out evaluation. Random splits over overlapping temporal windows leak, and a camera-hashed split can place an entire traffic regime in a single partition. Evaluating one held-out camera at a time, and reporting the fold mean with an interval, is the protocol that makes a cross-camera claim meaningful.

# CHAPTER 3

# PROBLEM STATEMENT & OBJECTIVES

## 3.1 Problem Statement

Given only the video stream of a camera already installed at a signalised urban intersection carrying heterogeneous, weakly lane-disciplined traffic, predict the congestion level of each of the four approaches 60 seconds ahead, and use that prediction to select signal phases so that mean vehicle waiting time is lower than under the best-configured Webster plan — without installing any additional sensor, without transmitting any raw frame off the capture device, and without the system failing unsafely when either the forecast or the network becomes unavailable.

The problem decomposes into four coupled difficulties. Counting is not detection: a detector that scores well on mAP can still mis-count under occlusion, and the operating point that maximises detection accuracy is not the one that minimises counting error. Labelling is not free: congestion has no ground-truth sensor here, so labels must be derived from counts under a stated, per-camera-calibrated rule. Forecasting is not description: the model must predict a state it cannot see. And evaluation is easily circular: labels derived from detector counts cannot be used to judge baselines that also consume detector counts, which is why the test split must be human-verified while training data may remain automatically labelled.

## 3.2 Objectives

The project pursues six objectives.

**O1.** Assemble a licence-audited corpus of Indian and comparable unstructured-traffic camera footage, and fine-tune a YOLOv8s detector over eight vehicle classes that include the auto-rickshaw, e-rickshaw and cattle absent from COCO.

**O2.** Derive per-lane congestion labels from detector counts using PCU-weighted occupancy with per-camera thresholds, so that a label means the same thing at a six-lane arterial and a two-lane approach.

**O3.** Design MFSTNet: a gated bidirectional cross-attention fusion of frozen CNN and ViT features, followed by recurrent and attention-based temporal modelling, predicting per-lane congestion 60 s ahead.

**O4.** Quantify the contribution of every architectural component through an eight-configuration ablation that includes a linear probe, evaluated leave-one-camera-out so that no reported score is a memorised camera.

**O5.** Train a PPO signal controller on a SUMO intersection whose observation vector carries the forecast, and measure it against a swept, best-configuration Webster baseline over 30 paired seeds with a pre-registered disqualification rule.

**O6.** Demonstrate an edge-to-server prototype in which no raw frame leaves the capture device, and in which both required fallback paths — forecast failure and network loss — are exercised.

## 3.3 Scope, Assumptions and Delimitations

The scope is a single four-way intersection observed by a single fixed camera, with four approaches and a two-phase signal plan. Lane geometry is placed by hand once per camera and reviewed, because automatically derived lane centres were measured to move by 0.164 of the frame width when the detector changed, which is large enough to reassign vehicles between approaches. The control policy is trained and evaluated in microsimulation; deployment on live signal hardware is explicitly out of scope for an academic project and would require authority approval and a safety case.

Two assumptions are stated rather than buried. The camera is assumed to be static, since the lane geometry is fixed at calibration. And the simulator is assumed to approximate heterogeneous traffic adequately once its sublane model is enabled; the residual gap between simulated and observed behaviour is declared in the results rather than treated as absent.

# CHAPTER 4

# PROPOSED METHODOLOGY

## 4.1 System Architecture

The system is organised into four subsystems that are developed independently and integrated late: a dataset and detection pipeline, the MFSTNet forecasting model, a reinforcement-learning signal controller, and an edge-to-server prototype that carries the first three into operation. Figure 4.1 shows the block diagram. The three tiers communicate only over the message contract shown; no tier reaches into another's internals, and the safety limits that clamp a signal command live outside the transport that carries it, so a command cannot become unsafe by arriving through a different path.

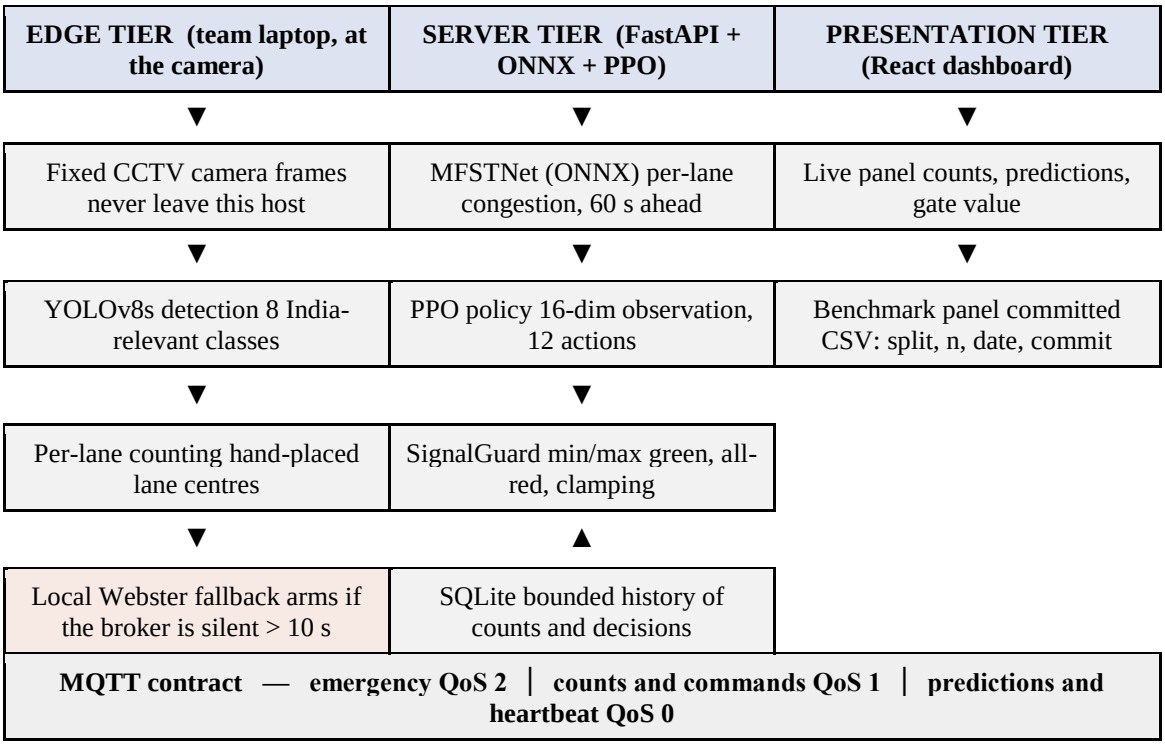



**Fig. 4.1 Block diagram of the proposed system**

The coupling that is easy to miss is that MFSTNet's output is not only a prediction to be displayed; it is part of the controller's observation. The observation vector is 16-dimensional and carries the four per-lane congestion predictions at indices 11 to 14. When the forecast is unavailable those four indices are zeroed and the vector keeps its length — shortening it would silently change the meaning of every later index and invalidate the trained policy.

## 4.2 Dataset and Corpus Construction

The dataset is built in two parts. Part A is curated from permissively licensed public sources; Part B contributes 1,500 to 3,000 annotated campus frames captured by the team. Every camera is declared in a manifest that records its location, viewpoint and how each property was determined — whether the footage was actually inspected or the property was inferred from a filename — so that a claim about coverage can be checked rather than asserted. Annotation covers eight classes: car, motorcycle, auto-rickshaw, e-rickshaw, bus, truck, bicycle and cattle.

The training corpus for the forecasting model is then constructed automatically from real footage. Figure 4.2 gives the pipeline. Clips are triaged for usability, lane centres are placed by hand and reviewed, the fine-tuned detector produces per-frame per-lane counts, and those counts are converted to congestion labels. The count file is retained separately from the label file, which means a threshold decision is a configuration change that re-derives labels in seconds rather than a reason to re-run the detector over hours of video.

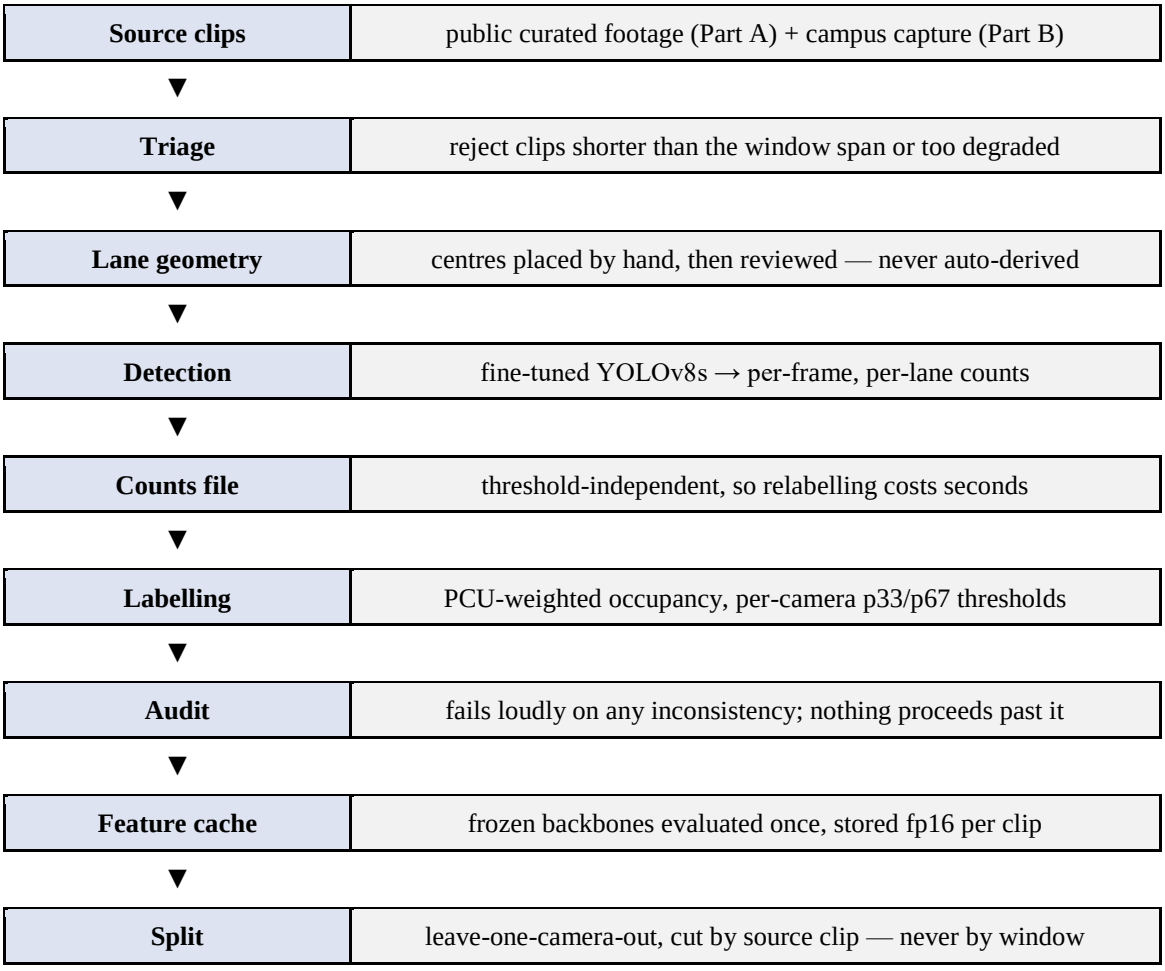



**Fig. 4.2 Corpus construction pipeline**

Each training sample is a window of T = 60 frames sampled every 2 s, labelled with the congestion state 60 s beyond the last observed frame, with successive windows starting 30 s apart. Because neighbouring windows share most of their frames, splits are cut by source clip and never by window; cutting by window would place near-duplicate samples on both sides of the split and inflate every score.

### 4.2.1 Congestion Labelling

A raw vehicle count is a poor congestion signal in mixed traffic, because a count of twenty two-wheelers and a count of twenty buses describe different roads. Labels are therefore defined on passenger-car-unit weighted occupancy using published Indian equivalence values, and the LOW/MEDIUM/HIGH thresholds are calibrated per camera at the 33rd and 67th percentiles of that camera's own distribution and recorded in the corpus manifest. An unmapped vehicle class raises an error rather than defaulting to a weight of one, so a silently mis-weighted class cannot reach the labels.

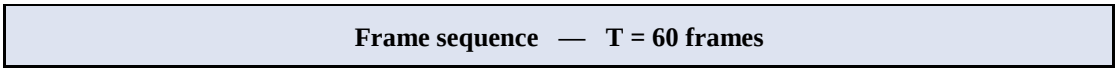
## 4.3 Vehicle Detection and Counting

YOLOv8s is fine-tuned on the joint dataset with a stratified 70/15/15 split. Two facts govern how it is used. First, detection accuracy and counting accuracy are separate quantities and are verified separately: the operating confidence threshold is chosen by minimising counting error against human-annotated frames, not by maximising mAP. Second, cross-camera generalisation is far weaker than within-camera performance, and it is the cross-camera number that governs deployment; both are reported.

Counts are attributed to approaches through hand-placed lane centres. Points that fall too far from every centre are left unassigned rather than snapped to the nearest one, because snapping fabricates a count in a lane that had none.

## 4.4 MFSTNet: Model Design

Figure 4.3 shows the model. Both backbones are frozen, so their outputs are computed once per clip and cached to disk as half-precision arrays; only the fusion, temporal and head parameters are trained. This is what makes an eight-configuration, four-fold, five-seed sweep tractable on a single 6 GB laptop GPU: the cache reduces the full ablation from an estimated 12.3 hours on CPU to roughly 1.5 hours. The cache is keyed by a hash covering backbone identity, input resize and normalisation, and a mismatch raises rather than warns, because a stale cache trains perfectly normally and reports wrong numbers.



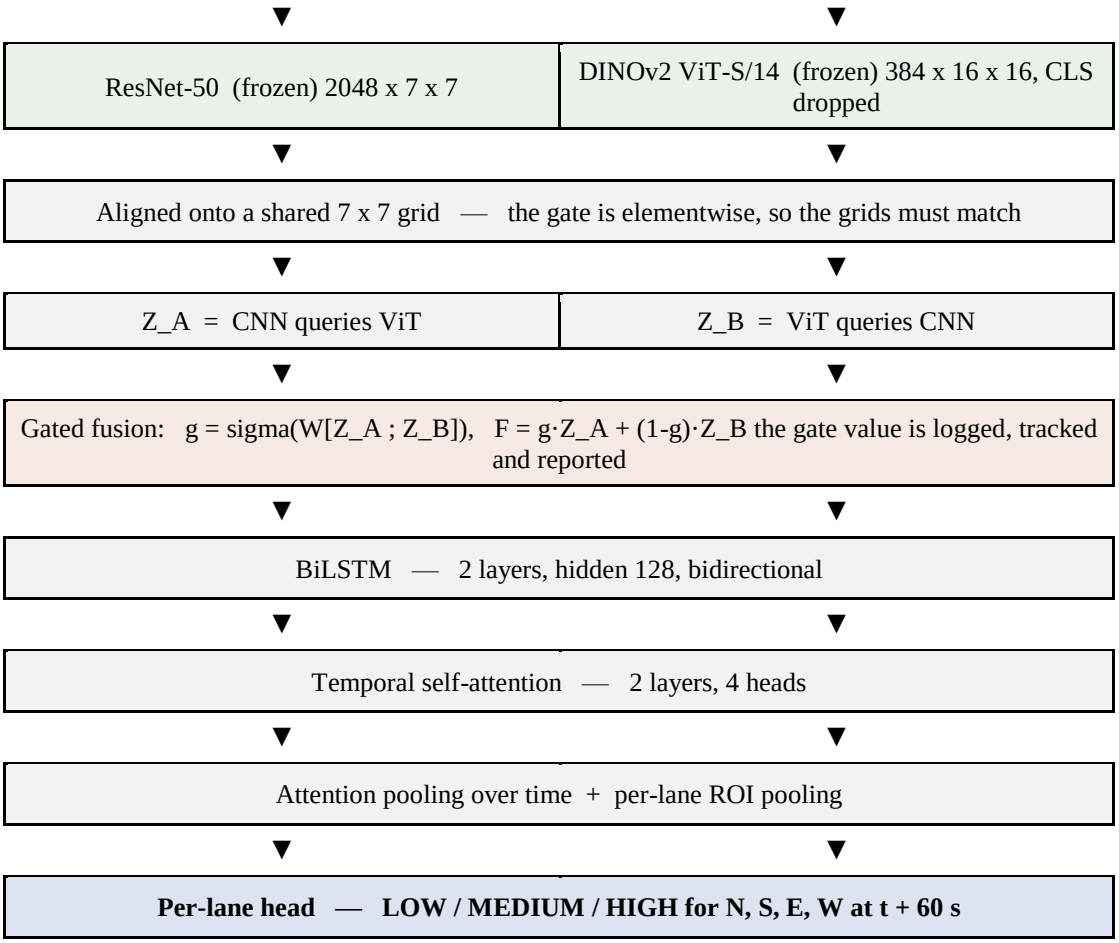



**Fig. 4.3 MFSTNet architecture**

### 4.4.1 Feature Extraction

ResNet-50 supplies a 2048-channel convolutional map and DINOv2 ViT-S/14 supplies 384-dimensional patch tokens. The two grids differ in size, and since the gate operates elementwise they must be aligned before fusion; both branches are therefore resampled onto a shared 7x7 grid, with the transformer's classification token dropped before reshaping.

### 4.4.2 Gated Bidirectional Cross-Attention

Two cross-attention passes are computed: Z_A, in which the convolutional features query the transformer features, and Z_B, in which the transformer features query the convolutional features. A learned gate then combines them, as in equations (4.1) and (4.2), where sigma is the logistic function, W is a learned projection and the semicolon denotes concatenation along the feature axis.

$$g = \sigma( W [ Z_A ; Z_B ] ) \quad (4.1)$$

$$F_{fused} = g \cdot Z_A + (1 - g) \cdot Z_B \quad (4.2)$$

The gate g is retained and reported. A gate that sits at 0.5 with negligible variance is telling the reader that the model is averaging the two branches rather than selecting between them,

which is an interpretable finding about the data and not a bug to be tuned away. Because it is a research artefact, the forward pass always returns it and no code path discards it.

### 4.4.3 Temporal Modelling

The fused per-frame representations form a sequence of length T, which a two-layer bidirectional LSTM with 128 hidden units per direction encodes into a 256-dimensional sequence. Two layers of four-head temporal self-attention then relate distant timesteps directly, and attention pooling collapses the sequence to a single vector — a learned weighting over time, in place of taking the last hidden state and discarding the rest of the window.

### 4.4.4 Per-Lane Prediction Head

A single globally pooled representation cannot produce four different per-lane answers, so pooling is per-lane: each approach's Voronoi region around its hand-placed centre defines a region of interest over the shared grid, and the head is applied to each region's pooled feature. This is why the feature cache must preserve spatial structure rather than store a single pooled vector per frame.

Training uses AdamW at a learning rate of 1e-4 with weight decay 1e-4, cosine annealing, a batch size of 32, up to 100 epochs with early stopping at patience 15, and cross-entropy weighted by inverse class frequency. Residency is managed by accumulating micro-batches to the specified batch size rather than by reducing it, since the batch size is a specified hyperparameter and available memory is not a reason to change one.

### 4.4.5 Ablation Design

Every component above is disableable by a configuration flag, giving the eight configurations in Table 4.1. No code is edited between configurations. Configuration H is the floor: frozen features, per-lane pooling, a mean over time, and a linear classifier. If H matches G, the architecture is unjustified on this data, and reporting that is the point of including it.

**Table 4.1 Ablation configurations**

| Config | Variant | What it enables or removes |
|---|---|---|
| A | CNN only | ResNet-50 features alone; no ViT, no fusion. |
| B | ViT only | DINOv2 ViT-S/14 features alone; no CNN, no fusion. |
| C | Concatenation | Both backbones, channel concatenation instead of attention. |

| Config | Variant | What it enables or removes |
|---|---|---|
| D | One-way cross-attention | CNN queries ViT only; the reverse path is disabled. |
| E | Bidirectional, no gate | Both directions, averaged rather than gated. |
| F | Gated + temporal attention | Adds temporal self-attention over the BiLSTM outputs. |
| G | Full model | Gated bidirectional fusion, temporal attention, attention pooling. |
| H | Linear probe | Frozen features, mean over time, linear classifier. The floor. |

## 4.5 Reinforcement-Learning Signal Controller

The controller is a PPO agent acting on a four-way intersection modelled in SUMO and driven through TraCI, with the sublane model enabled so that lateral filtering is representable. The action space is 12 discrete actions, being the choice of phase (north-south or east-west) crossed with six green durations of 10, 20, 30, 45, 60 and 90 seconds. The observation is the 16-dimensional vector described in Section 4.1. The reward, equation (4.3), is the reduction in total lane waiting time between successive decisions, penalised for starving an approach and lightly penalised for switching, so that the agent is discouraged from churning without being forbidden to adapt.

$$r_t = -(W_t - W_{t-1}) / 100 - 1.0 * n_{starved} - 0.05 * 1[switch] \quad (4.3)$$

Training uses a learning rate of 3e-4, 2048 steps per update, a minibatch of 64, a discount of 0.99, GAE lambda 0.95, a clip range of 0.2, an entropy coefficient of 0.01, and 500,000 timesteps. Minimum and maximum green times of 10 and 90 seconds and a 3-second all-red interval are enforced by actuation rather than learned, because a safety constraint that the agent could trade away is not a safety constraint.

The baseline deserves as much care as the policy. Webster's optimum cycle is sensitive to an assumed saturation flow whose published values for heterogeneous traffic span from 525 to 1283 PCU per hour per metre of approach width. Rather than choose one, the baseline sweeps the published range and the best-performing configuration becomes the baseline the learned policy must beat. Cycle clamping is logged and the clamp rate reported alongside the result: a Webster controller that clamps in every cycle has stopped computing anything and is a fixed-time plan under another name, and comparing against it would be misleading.

## 4.6 Edge-to-Server Prototype

The edge node runs detection and counting beside the camera and publishes only counts and predictions over MQTT; the control server hosts the policy and the exported model and returns signal commands. Quality of service differs by topic and is attached to the topic definition rather than accepted from the caller, so a caller cannot downgrade an emergency message: emergency traffic is QoS 2, counts and commands QoS 1, and predictions and heartbeats QoS 0. Two fallback paths are required behaviour. If model inference fails, the controller continues on raw counts with the forecast indices zeroed; if the broker is unreachable for more than 10 seconds, the edge node switches to a locally embedded Webster controller whose parameters were embedded at deployment time, since a fallback that activates because the network is down cannot fetch its own configuration.

The edge device for this project is a team laptop rather than an embedded accelerator, which is a budget decision and not a design one: the control logic and the message contract are unchanged. Every latency figure therefore states the host it was measured on, and laptop figures are labelled as proxy measurements.

## 4.7 Evaluation Protocol

The forecasting model is evaluated leave-one-camera-out over the cameras contributing at least ten windows. Each fold holds out one camera entirely for testing, with the remainder in training and validation, so a reported score cannot be the model recognising a camera it has already seen. The headline figure is the mean across folds with a bootstrap 95% confidence interval computed by the percentile method over 10,000 resamples [16], quoted together with the number of folds, because an interval over four points is not an interval over thirty.

Every configuration is run at five seeds. A single-seed table ranks initialisations rather than architectures: on this corpus one configuration moved by more than 0.24 macro-F1 across seeds on a single fold, which is larger than the gap between most pairs of configurations. The evaluation harness therefore reports the seed spread beside every mean and warns when the spread exceeds the closest gap between two configurations, since those two are then not separated by the experiment.

Metrics are macro-F1 as the headline, with accuracy, weighted F1, quadratic weighted kappa and an ordinal error rate reported alongside, because congestion classes are ordered and confusing LOW with HIGH is not the same error as confusing LOW with MEDIUM. Control results are reported as mean vehicle waiting time over 30 paired evaluation seeds

per method, with a bootstrap 95% confidence interval, a paired t-test at alpha = 0.05 and Cohen's d. Pre-registered disqualification rules apply: a method that completes fewer than 85% of trips, or a Webster arm that clamps in more than half of its cycles, is disqualified from the comparison and reported as such rather than quietly quoted.

All raw per-run results are committed as CSV, not only summary tables, and every table is generated from those CSVs by a committed script. Seeds are fixed across PyTorch, NumPy and the RL library, and a machine-checked claims file holds each scope claim as the sentence it would appear as in the report, paired with an assertion that is run against the corpus. A claim whose assertion fails blocks table generation; there is no override.

## 4.8 Preliminary Results

The pipeline is implemented end to end and has produced its first measurements on a pilot corpus. They are reported here in full, including the ones that do not support the vision hypothesis, because a synopsis that reports only the favourable half would misrepresent the state of the work. Table 4.2 gives the ablation across the eight configurations, and Table 4.3 gives the count-only baselines evaluated on the identical folds and seeds.

**Table 4.2 Leave-one-camera-out ablation results (4 folds x 5 seeds, 20 runs per configuration)**

| Config | Macro-F1 [95% CI] | Accuracy [95% CI] | Seed spread |
|---|---|---|---|
| A | 0.3596 [0.3144, 0.4051] | 0.5318 [0.4761, 0.5809] | 0.1604 |
| B | 0.4059 [0.3607, 0.4491] | 0.5624 [0.4887, 0.6302] | 0.1931 |
| C | 0.4405 [0.3724, 0.5124] | 0.5996 [0.5317, 0.6627] | 0.1915 |
| D | 0.2787 [0.2094, 0.3475] | 0.4281 [0.3325, 0.5172] | 0.1247 |
| E | 0.4123 [0.3333, 0.4870] | 0.5777 [0.4807, 0.6632] | 0.2090 |
| F | 0.3900 [0.3077, 0.4675] | 0.5429 [0.4407, 0.6329] | 0.2136 |
| G | 0.3950 [0.3226, 0.4642] | 0.5513 [0.4537, 0.6380] | 0.1984 |
| H | 0.4155 [0.3797, 0.4494] | 0.5914 [0.5423, 0.6385] | 0.1649 |

**Table 4.3 Count-only baselines on the same folds and seeds**

| Count-sequence baseline | Macro-F1 | Accuracy |
|---|---|---|
| gru counts | 0.5663 | 0.7188 |
| naive last value | 0.5571 | 0.7110 |
| lstm counts | 0.5405 | 0.6955 |
| xgboost counts | 0.4691 | 0.6837 |

Three readings follow, and each is stated as the evidence supports it. First, the strongest vision configuration is C at a macro-F1 of 0.4405 [0.3724, 0.5124], while the strongest count-only baseline (gru counts) reaches 0.5663 on the same folds and seeds. On this corpus the appearance features are therefore not yet earning their place, and the vision claim is reported as open.

Second, the full model and the linear probe cannot be separated. Across 4 folds the mean difference between them is -0.0204 with a 95% interval of [-0.1544, 0.0657], against a minimum detectable difference of 0.2115 at this fold count. The verdict recorded by the analysis script is “cannot distinguish”, which is a statement about the power of a four-fold experiment as much as about the architecture.

That mean requires its composition stated, because it is not evenly made. Fold by fold the difference is 0:+0.0237 1:-0.2163 2:+0.0796 3:+0.0313: the full model is ahead on three of the four folds and loses the fourth by a margin larger than the other three gains combined. That fourth fold holds out the M6 Motorway, the one camera declared “structured” against a target of “mixed”, and it contributes 41.2% of the corpus windows and 52.6% of every HIGH label. Averaged over the three in-regime folds alone the sign reverses and the full model leads by 0.0449; the reversal survives scoring only the classes present in each fold, so it is a property of which camera is held out rather than of which classes were evaluable. Neither aggregate corrects the other — one measures transfer to an out-of-regime camera and one measures generalisation within the target regime — and this report quotes the four-fold figure as its headline because that is the protocol registered in advance. The gate itself sits at 0.5299 with a standard deviation of 0.0090: the model is averaging its two branches, not selecting between them.

Third, the control result is positive and holds up. Against the swept, best-configuration Webster baseline at 14.053 s mean waiting time, the learned policy achieves 11.334 s over 30 paired seeds — an improvement of 19.35% with $p < 0.0001$ and Cohen's $d = -1.34$. This arm carries no forecast, so it measures the value of learned phase selection alone; the contribution of the forecast is a separate experiment and is not claimed here.

The corpus is also honest about what it does not yet contain. No camera currently eligible for a fold is Indian, and the HIGH class is scarce in three of the four folds. Both conditions are encoded as failing assertions in the machine-checked claims file, which is why report-table generation is currently blocked; expanding the corpus with Part B campus footage is the work that clears them, and no claim about the Indian domain is made until it does.

One limitation is sharp enough to state on its own, because it bounds every number in this report. None of the four fold-eligible cameras is Indian. Three are Bangladeshi and one is a British motorway; the two Indian sources in the corpus contribute six and three windows, which is below the ten-window threshold a camera needs to be scored as a fold. Every result here is therefore evidence about generalisation across cameras in heterogeneous South Asian traffic, and is not evidence about generalisation to an Indian camera. That is a limitation of the evidence and not a reason to withhold it: the out-of-regime fold is the only external-domain test the project has, and removing it to improve an average would be the selective reporting this project's own rules forbid. A further consequence follows for one class in particular — the camera held out in fold 0 contains no HIGH windows at all, so that fold cannot evaluate the HIGH class, and its score must not be read as evidence either for or against performance on it.

# CHAPTER 5

# HARDWARE AND SOFTWARE REQUIREMENTS

The requirements below are the ones this project actually uses; nothing is listed because it is conventional. Two entries are worth their explanation. Training runs locally on a 6 GB laptop GPU rather than on a cloud instance, which is possible only because the frozen backbones are evaluated once and cached — an uncached step at the specified batch size would push 1,920 frames through both backbones and would not fit. And the database is SQLite rather than a separate server, because the prototype stores a bounded history of counts and decisions and a database server would be infrastructure without a purpose. Table 5.1 lists the hardware and Table 5.2 the software.

## 5.1 Hardware Requirements

**Table 5.1 Hardware requirements**

| Component | Specification | Purpose in this project |
|---|---|---|
| Training workstation | Laptop, 8-core x86-64 CPU, 16 GB RAM | Feature caching, MFSTNet training, ablation sweeps. |
| GPU | NVIDIA RTX 4050 Laptop, 6 GB VRAM, compute capability 8.9 | Frozen-backbone feature extraction and training; bf16/fp16 mixed precision. |
| Storage | 512 GB SSD, of which roughly 60 GB for video, cache and weights | Source clips, the memory-mapped feature cache and model checkpoints. |
| Edge node | Team laptop acting as the edge device (ADR-003) | Detection, counting and the local fallback controller. Every latency figure states this host, because it is not a Jetson. |
| Camera | Existing municipal or campus CCTV stream, 720p or better | Frame source. Frames are processed in place and never transmitted. |
| Network | Wi-Fi or Ethernet LAN; MQTT broker within 200 ms | Carries counts, predictions and signal commands only. |

## 5.2 Software Requirements

**Table 5.2 Software requirements**

| Category | Software and pinned versions |
|---|---|
| Operating system and language | Windows 11 and Ubuntu 22.04 (both supported); Python 3.12+ |

| Category | Software and pinned versions |
|---|---|
| Deep learning | PyTorch 2.13.0 with CUDA 12.6; torchvision 0.28.0; timm 1.0.28 for the DINOv2 backbone; Ultralytics YOLOv8 8.4.118 |
| Reinforcement learning and simulation | Stable-Baselines3 2.9.0; Gymnasium 1.3.0; Eclipse SUMO 1.27.1 with TraCI and sumolib |
| Serving and deployment | ONNX 1.22.0 and ONNX Runtime 1.29.0; FastAPI 0.141.1 with Pydantic 2.13.4; Uvicorn 0.52.4; Eclipse Mosquitto with Paho-MQTT |
| Data, storage and analysis | SQLite (embedded; no separate database server); NumPy 2.5.2, pandas 3.0.5, SciPy 1.18.0; scikit-learn 1.9.0, XGBoost 3.4.1; OpenCV 5.0.0 |
| Interface | React 18 with Vite and Zustand; Recharts and native WebSocket |
| Engineering tooling | Git with Git LFS for model weights; pytest 9.1.1 with GitHub Actions CI; MLflow and TensorBoard for experiment tracking; Docker Compose |

Versions are pinned in a committed requirements file, which is the authority; any version written in prose is a summary of it. The CUDA-enabled build of the deep-learning framework must be installed from its own package index before the remaining packages, because the default index resolves to a CPU-only wheel that then fails at run time rather than at install time.

# CHAPTER 6

# EXPECTED OUTCOMES

The project is expected to deliver seven artefacts, each of which is measurable rather than descriptive. Table 6.1 lists them with the form each one takes on delivery.

**Table 6.1 Expected deliverables**

| Deliverable | Description |
|---|---|
| Curated dataset | Part A curated from permissively licensed public sources plus Part B of 1,500-3,000 annotated campus frames, with a per-camera provenance record. |
| Fine-tuned detector | YOLOv8s weights over eight India-relevant classes, with the counting operating point calibrated against human-verified counts. |
| MFSTNet model | Trained weights, an ONNX export for CPU inference, and the exposed gate value as a research artefact rather than an internal detail. |
| Ablation and evaluation results | Per-run CSVs for all eight configurations across every fold and seed, with bootstrap confidence intervals generated by a committed script. |
| PPO controller | A trained policy plus a 30-seed paired benchmark against Webster and fixed-time control across three demand regimes. |
| Working prototype | Edge node, MQTT contract, FastAPI control server with safety clamping outside the transport layer, and a two-panel monitoring dashboard. |
| Reproducibility package | Pinned requirements, fixed seeds, committed raw results and a public code release, so that every reported number can be regenerated. |

## 6.1 Measurable Targets

The following targets are the criteria against which the finished work will be judged. They are stated in advance so that a result that misses them is visible as a miss rather than reframed after the fact.

1. Detection: at least 0.85 mAP@0.5 on a held-out test split of the curated dataset, with cross-camera performance reported separately and without averaging the two.
2. Counting: mean absolute per-frame counting error below 1.5 vehicles at the chosen operating point, verified against human-annotated frames rather than against detector output.

3. Forecasting: the full model to exceed the linear probe by a margin larger than the seed spread, on a corpus large enough for that margin to be detectable. Where it does not, the finding is reported as an inconclusive comparison with its minimum detectable difference stated.
4. Control: a statistically significant reduction in mean vehicle waiting time against the best-configured Webster baseline over 30 paired seeds, with the baseline's clamp rate disclosed. The preliminary benchmark already meets this at 19.35%.
5. Latency: model inference within 150 ms on server CPU through the exported graph, a control decision within 50 ms, and message delivery within 200 ms, each figure labelled with the host it was measured on.
6. Reproducibility: every reported number regenerable from committed code, committed raw results and pinned dependencies on a clean machine, with seeds fixed and documented.

## 6.2 Academic Outcome

The intended academic output is a conference submission. The defensible contribution is not a new fusion mechanism — those are published — but the combination of a camera-only forecast-to-control path, the gate reported as an artefact rather than hidden as an activation, and a leave-one-camera-out evaluation reported with the linear-probe floor beside it. A negative or inconclusive vision result, reported with its statistical power, remains a publishable contribution under that framing.

# CHAPTER 7

# CONCLUSION

This synopsis has proposed MFSTNet, a camera-only system that forecasts per-lane congestion 60 seconds ahead at a signalised urban intersection and uses that forecast to drive a reinforcement-learning signal controller. The proposal is built around infrastructure that Indian cities already own — fixed cameras — and requires no additional roadside sensing, no raw video to leave the point of capture, and no unsafe behaviour when either the forecast or the network fails.

The methodology has been described in full: a licence-audited two-part dataset with eight India-relevant vehicle classes; PCU-weighted, per-camera-calibrated congestion labels derived from detector counts; a gated bidirectional fusion of frozen convolutional and self-supervised transformer features with per-lane region pooling and temporal attention; a PPO controller whose observation carries the forecast in a fixed 16-dimensional contract; and an edge-to-server prototype with two required fallback paths.

What distinguishes the plan is the evaluation rather than the architecture. Every module is disableable by configuration so that an eight-way ablation runs without a code edit; the linear probe is included as the floor that a frozen-backbone study owes its reader; evaluation is leave-one-camera-out so that no score is a memorised camera; and all results are reported as means with bootstrap confidence intervals across folds and seeds, with raw per-run data committed alongside the code.

Preliminary results are consistent with that discipline. The control half is already positive: a 19.35% reduction in mean waiting time against a swept Webster baseline over 30 paired seeds, at $p < 0.0001$. The vision half is not yet: the best configuration reaches 0.4405 macro-F1 against 0.5663 for a count-only sequence model, and the full model cannot presently be separated from the linear probe at four folds. Those findings are reported rather than deferred, and they set the remaining work: expand the corpus with Indian campus footage so that the domain claim can be made honestly, raise the number of evaluable cameras so that the architecture comparison has the power to resolve, and complete the forecast-carrying control arm so that the value of the forecast is measured rather than assumed.

The expected significance is a low-cost, privacy-preserving adaptive signal control path for unstructured urban traffic, delivered together with the evidence needed to judge whether each of its parts is earning its place.

# REFERENCES


[1] A. Vaswani et al., "Attention is all you need," in Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), 2017, pp. 5998-6008.

[2] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," in Proc. Int. Conf. Learn. Represent. (ICLR), 2021.

[3] M. Oquab et al., "DINOv2: Learning robust visual features without supervision," Trans. Mach. Learn. Res., 2024.

[4] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), 2016, pp. 770-778.

[5] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLOv8," 2023. [Online]. Available: https://github.com/ultralytics/ultralytics

[6] G. Varma, A. Subramanian, A. Namboodiri, M. Chandraker, and C. V. Jawahar, "IDD: A dataset for exploring problems of autonomous navigation in unconstrained environments," in Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV), 2019, pp. 1743-1751.

[7] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," arXiv:1707.06347, 2017.

[8] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, and N. Dormann, "Stable-Baselines3: Reliable reinforcement learning implementations," J. Mach. Learn. Res., vol. 22, no. 268, pp. 1-8, 2021.

[9] P. A. Lopez et al., "Microscopic traffic simulation using SUMO," in Proc. IEEE Int. Conf. Intell. Transp. Syst. (ITSC), 2018, pp. 2575-2582.

[10] F. V. Webster, "Traffic signal settings," Road Research Technical Paper No. 39, Road Research Laboratory, London, U.K., 1958.

[11] Indian Roads Congress, IRC:SP:41-1994, Guidelines for the Design of At-Grade Intersections in Rural and Urban Areas. New Delhi, India: IRC, 1994.

[12] Indo-HCM, Indian Highway Capacity Manual. New Delhi, India: CSIR-Central Road Research Institute, 2017.

[13] H. Wei, G. Zheng, V. Gayah, and Z. Li, "Recent advances in reinforcement learning for traffic signal control: A survey of models and evaluation," ACM SIGKDD Explorations Newsletter, vol. 22, no. 2, pp. 12-18, 2021.

[14] S. Hochreiter and J. Schmidhuber, "Long short-term memory," Neural Computation, vol. 9, no. 8, pp. 1735-1780, 1997.

[15] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in Proc. ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining, 2016, pp. 785-794.

[16] B. Efron and R. Tibshirani, An Introduction to the Bootstrap. New York, NY, USA: Chapman & Hall, 1993.